

Technology Stack

Date	27 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript	Used to create a responsive and user-friendly web interface where farmers can enter soil and environmental parameters and receive crop recommendations instantly.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, Flask	Python provides extensive machine learning support, while Flask offers a lightweight web framework for integrating the trained ML model with the web application and processing prediction requests efficiently.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	Crop_recommendation.csv Dataset, Joblib (.joblib.pkl) Model Files	The CSV dataset is used to train the machine learning model, while Joblib stores the trained model and scaler for fast and efficient predictions without retraining.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Flask Local Server (Current), Docker & Render/AWS (Future)	The application currently runs on a local Flask server for development and testing. The architecture is designed to support future deployment on cloud platforms such as Render or AWS using Docker containers.
5	Version Control & CI/CD	Git, GitHub	Git provides version control for tracking code changes and collaboration, while GitHub serves as the central repository and supports future CI/CD automation using GitHub Actions.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib (Current) OpenWeatherMap API	Scikit-learn is used for training and evaluating machine learning models. Pandas and NumPy handle data preprocessing and analysis, Matplotlib

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		(Future Enhancement)	and Seaborn provide visualization, and Joblib is used to save and load the trained model. In future versions, the OpenWeatherMap API can automatically fetch real-time weather data to improve crop recommendation accuracy.